Results and Analysis

Disguising AI-Generated Text as Human Written

Overview

This document presents the experimental results of different techniques evaluated for improving the human-likeness of AI-generated text. The approaches compared include Text Generation, Adversarial Search, Retrieval-Augmented Generation (RAG), Back-Translation, and GAN-based refinement.

Detector Score Comparison

Table 1: Comparison of Detectability Scores

Method	**Initial Score**	**Final Score**	**Improvement**
Baseline (No Enhancement)	0.0269	–	–
RAG	0.0269	0.0049	∼81.8% reduction
Adversarial Search	∼0.999	∼0.0009	Significant reduction
GAN-based Approach	High	Lower	Moderate improvement
Back-Translation	Moderate	Lower	Noticeable improvement

Qualitative Comparison

Table 2: Comparison of Different Techniques

Method	**Human-likeness**	**Diversity**	**Detectability Reduction**
Text Generation	Medium	Medium	Low
RAG	High	Medium	Medium
Adversarial Search	High	High	Very High
Back-Translation	Medium	High	Medium
GAN-based Approach	High	High	Medium

Sample Outputs

Text Generation Output

```
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EVALUATION REPORT: submission.txt
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📊 Binoculars Score: 0.9957
   → HUMAN-LIKE

📉 Perplexity (GPT-2): 35.32
   → Lower = more AI-like

🔢 Avg Token Rank: 162.25
   → Lower = more predictable (AI-like)

📊 GLTR Distribution:
   top10: 65.89%
   top100: 16.94%
   top1000: 12.76%
   rest: 4.41%

🤖 RoBERTa Detector:
   Label: Real
   Confidence: 0.9703

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🧠 Final Composite Score: 1.4489
🎯 FINAL VERDICT: LIKELY HUMAN
```

Figure 1: Sample Output from Base Text Generation

Adversarial Search Output

```
Generating for topic 030
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=900) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=900) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=900) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta
Initial score: 0.9996614456176758
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=900) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta
Final score: 0.0009104351047426462
```

Figure 2: Refined Output after Adversarial Search

RAG Output

```
Passing `generation_config` together with generation-related arguments=({'max_new_tokens', 'top_p', 'do_sample', 'temperature'}) is deprecated
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=600) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta

Generating BASELINE...
Setting `pad_token_id` to `eos_token_id`:2 for open-end generation.
Both `max_new_tokens` (=600) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documenta
Baseline Score: 0.02698993682861328

Generating RAG...
RAG Score: 0.004931807518005371

Comparison:
Baseline: 0.02698993682861328
RAG: 0.004931807518005371
```

Figure 3: Output with Retrieval-Augmented Context

Back-Translation Output

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📊 Binoculars Score: 0.9997
   → HUMAN-LIKE

📉 Perplexity (GPT-2): 228.66
   → Lower = more AI-like

🔢 Avg Token Rank: 412.71
   → Lower = more predictable (AI-like)

📊 GLTR Distribution:
   top10: 40.43%
   top100: 26.95%
   top1000: 21.28%
   rest: 11.35%

🤖 RoBERTa Detector:
   Label: Fake
   Confidence: 0.6486

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🧠 Final Composite Score: 1.4593
⚠️FINAL VERDICT: POSSIBLE AI
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```

Figure 4: Output after Back-Translation

GAN-based Output

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EVALUATION REPORT: submission.txt
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📊 Binoculars Score: 0.9913
   → HUMAN-LIKE

📉 Perplexity (GPT-2): 72.65
   → Lower = more AI-like

🔢 Avg Token Rank: 363.57
   → Lower = more predictable (AI-like)

📊 GLTR Distribution:
   top10: 57.22%
   top100: 18.56%
   top1000: 15.44%
   rest: 8.78%

🤖 RoBERTa Detector:
   Label: Real
   Confidence: 0.9995

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🧮 Final Composite Score: 1.5757
🎯 FINAL VERDICT: LIKELY HUMAN
```

Figure 5: Output after GAN-based Refinement

Summary

- Adversarial search achieves the lowest detectability scores.
- RAG improves contextual grounding and realism.
- Back-translation introduces variability and reduces repetitive patterns.
- GAN-based refinement enhances stylistic richness.
- No single method is universally optimal; each method contributes differently.